STARTUP SUCCESS PREDICTION USING SUPERVISED LEARNING MODELS

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INTRODUCTION

Project Context:

Startups are key drivers of innovation and economic growth, introducing disruptive ideas and shaping the future of industries. However, the road to success is fraught with challenges, and research indicates that approximately 90% of startups fail within their first few years. This highlights the importance of understanding the factors that determine a startup's success or failure.

In today's fast-paced and competitive environment, entrepreneurs and investors often lack effective tools to assess a startup's potential viability. Many decisions rely on limited data or intuition, increasing the risks of failure and resource misallocation.

Problem Statement:

The problem this project addresses is the difficulty in reliably predicting whether a startup will succeed or fail. While other tools may provide partial insights into market fit, funding, or operational challenges, they lack a comprehensive approach to determining overall success probability.

This challenge offers an opportunity to leverage machine learning to analyze historical data and build a system that predicts a startup’s outcome—success or failure—based on key factors.

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Project Goals:

This project focuses on :

- Building predictive models using supervised machine learning techniques :
  - K-Nearest Neighbors (KNN)
  - Logistic Regression
  - Naive Bayes
  - Decision Tree
- Evaluating model performance to identify the most accurate approach
- Providing actionable insights into the key factors driving startup success

Target Audience:

The primary beneficiaries of this project are:

- Entrepreneurs: By predicting the likelihood of success or failure, they can better evaluate their projects and decide whether to proceed or pivot.
- Investors: They gain insights to identify potential risks and make informed decisions about funding startups.
- Incubators and Accelerators: They can use the tool to prioritize startups with a higher probability of success for their programs.

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PLANNING AND METHODOLOGY

To achieve the main goal of predicting whether a startup will succeed or fail, we have adopted a supervised learning approach. This approach uses machine learning algorithms to train predictive models on a labeled dataset. Below are the key steps of our project :

1. Selection of Supervised Learning Models:

We selected four algorithms for comparison based on their strengths and suitability for classification tasks:

- **k-Nearest Neighbors (kNN):** Distance-based algorithm for classification.
- **Logistic Regression:** A statistical model efficient for binary outcomes (success or failure).
- **Naive Bayes:** A probabilistic classifier
- **Decision Tree:** A tree-based model that splits data based on decision rules, providing a clear structure

2. Data Collection:

For our project, we selected a dataset from Kaggle that contains comprehensive information about startups from different countries and fields. This dataset is valuable as it includes various features that help in analyzing the factors affecting the success or failure of startups.

Below is a detailed overview of the dataset:

a. **Description:** The dataset includes data about startups across different sectors, regions, and funding statuses. It provides insights into company characteristics and the factors that might influence their success or failure.
b. **Link:** https://www.kaggle.com/datasets/yanmaksi/big-startup-secsees-fail-dataset-from-crunchbase/data
c. **Columns (Features):**
   - **permalink:** A unique URL path associated with the startup.
   - **name:** The name of the startup.
   - **homepage_url:** URL of the startup's website.
   - **category_list:** Industry or field of the company (e.g., software, biotechnology).
   - **funding_total_usd:** Total funding received by the startup in USD.
   - **status:** The current operating status of the startup (e.g., operating, closed).
   - **country_code**: The country code where the startup is located.
   - **state_code**: The state code for the company's location.
   - **region:** The region where the startup operates (e.g., SF Bay Area, USA).
   - **city:** The city where the company is located.
   - **funding_rounds:** the number of rounds of funding the startup has gone through.
   - **founded_at:** The date when the startup was founded.
   - **first_funding_at:** The date when the startup raised its first round of funding. This marks the initial investment in the startup.
   - **last_funding_at:** the date when the startup raised its most recent round of funding.

3. Data Preprocessing:

a. **Data Cleaning:** Handle missing values(funding_total_usd, country_code, founded_at) and standardize formats.
b. **Encoding Data**: Convert categorical variables (status, category, country, ..) into numerical data for model training.
c. **Normalization:** Scale numerical data to improve the performance of algorithms(funding_total_usd).

4. Model Training and Validation:

a. **Data Splitting:** Split the data into training and testing sets (80% for training, 20% for testing).
b. **Train and Test the model**
c. **Evaluation:** Use of performance metrics such as accuracy and confusion_matrix.

5. Models Performance Comparison:

After training, we compared the performance of the models to identify the best-performing algorithm for our prediction task :

a. Accuracy of KNN: 0.78
b. Accuracy of Logistic Regression: 0.79
c. Accuracy of Naive Bayes: 0.14
d. Accuracy of Decision Tree: 0.74

6. Results and Key Insights:

- **Data Independence Assumption in Naive Bayes:** Naive Bayes classifiers make a critical assumption that the features in the dataset are independent of each other. In the context of predicting startup success or failure, this assumption can be problematic. Key factors that influence the success of a startup, such as funding, year of foundation, and location, are often correlated. For instance, a startup located in a major tech hub may have access to more funding and resources than a startup in a less developed region.
- The independence assumption in Naive Bayes may fail to capture these interdependencies between variables, which results in the model not leveraging important contextual relationships and ultimately lowering its prediction accuracy.
- **Complexity of the Dataset:** KNN, Logistic Regression and Decision Trees are more flexible in handling complex relationships between features. KNN can capture non-linear relationships by considering proximity, Logistic Regression can model interactions between features especially with regularization, and Decision Trees can capture hierarchical, non-linear patterns through splits on features.
- **Data Quality and Size:** With a large, well-cleaned dataset, models like KNN, Logistic Regression and Decision Trees can all perform well because they have more data to learn complex patterns. Naive Bayes, on the other hand, is relatively simple and may not be powerful enough for complex datasets, especially when there are feature dependencies.

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- **Features Importance with Decision Trees:** Decision Trees are particularly useful for understanding feature importance because they can assign a score to each feature based on how important it is in making predictions.
  - **funding_total_usd :** This feature has the highest importance score ≈ 0.264, indicating that the total funding amount a startup has received is the most influential factor in predicting startup success.
  - **first_funding_at** : The first funding date also plays a significant role ≈ 0.127. This indicates that when a startup receives its first funding may impact its success, potentially suggesting that startups receiving early investments have higher chances of success.
  - **founded_at :** The founding date has a moderate level of importance ≈ 0.112, which may imply that the age of a startup could influence its likelihood of success, possibly because older companies have more time to establish their products, markets, and customer bases.
  - **Country and category features** have very low importance scores. This could indicate that factors like funding and timing are more crucial than the startup's geographical location or industry sector.